Lab 4: Statements

You added an expression interpreter in the midterm. In this lab, you will extend your language to support expression statements and print statements. If you have not finished the previous labs, now is the time to catch up.

Tasks

You will submit both your **code** and a **report** for this lab. Partial credit will be awarded for all tasks.

Report quality and submission correctness together account for at most **10% of the total score**, included within the task weights below.

- **Statements (5 points):**
  Parse and execute print statements and expression statements in your language.
- **Panic-Mode Error Recovery (5 points):** Implement panic-mode error recovery so that the parser can report a syntax error and continue parsing subsequent statements.

Testing and Report

Use the equivalent of the following two test cases in your language:

```
"one"
true;
print 2 +
```

and

```
4 + 2;
print "one";
print true;
print 1 + 2;
print nil;
```

Your report should include the output of both test cases, including any syntax error messages. Place your report under `report/` in your repository.

Submission Instructions

Submit a **link to the exact commit in your GitHub repository** on Blackboard, using the format `https://github.com/<owner>/<repo>/commit/<commit-hash>`.

The submitted commit must include:

- All implementation files in `src/`, including the parser and reused scanner, AST classes, and AST printer.
- All test inputs for this lab under `test/`.
- The lab report in `report/`.

Use the following repository structure:

```
repository/
├── src/                # Implementation files
├── test/
│   └── lab*/           # Lab * test inputs and test scripts
└── report/
    └── lab*.md         # Lab * report, including setup and run
instructions
```

Ensure that the instructor can access the repository and reproduce your results from the submitted commit.

Late Penalty

Submit your lab on time. Submitting one lab late can reduce the time available to complete subsequent labs. All labs are due at 11:59 p.m. Eastern Time on the specified due date. A **10% penalty applies for each day late, up to 50%**. Submissions more than five days late receive a score of zero.

AI Policy

Using AI assistance for the labs is ALLOWED but discouraged. You must clearly mark any AI-generated code.

Mark it like this, using the appropriate comment syntax for your language:

```
// --- start AI code ---

// --- end AI code ---
```

Using AI-generated code without marking it is considered academic misconduct and will result in a score of zero for the lab. Mistakes in AI-generated code will result in double the point deductions specified in the rubric.

Reuse of code from previous labs is ALLOWED and encouraged.